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MARWADI UNIVERSITY

Faculty of Technology

Information and Communication Technology

B.Tech **SEM: VI** **Mid 2 Exam** **April 2024**

Subject: - Artificial Intelligence (01CT0616) **Total Marks:-30**

1. What is Artificial Intelligence?

A) Database. B) Simulation of human intelligence

C) Hardware. D) Network **Answer:** B

2. What does NLP stand for?

A) Network Layer Protocol B) Natural Language Processing

C) Neural Language Program D) Numeric Logic Processing **Answer:** B

3. What is Machine Learning?

A) Manual coding B) Learning from data

C) Hardware design D) Networking **Answer:** B

4. What is a dataset?

A) Single value B) Collection of data

C) Software D) Tool **Answer:** B

5. What is tokenization?

A) Encryption B) Splitting text

C) Storage D) Compression **Answer:** B

6. Which of the following best describes supervised learning in Artificial Intelligence?

A) Learning with labeled data B) Learning without labels

C) Learning using rewards D) Learning randomly **Answer:** A

7. Which technique is commonly used to measure similarity between two text documents in NLP applications?

A) Sorting B) Cosine Similarity

C) Searching D) Hashing **Answer:** B

8. What is the purpose of TF-IDF in text processing and information retrieval systems?

A) Image processing B) Keyword extraction

C) Data storage D) Encryption **Answer:** B

9. Which of the following correctly defines reinforcement learning in Artificial Intelligence systems?

A) Learning with labeled data B) Learning through rewards and penalties

C) Learning without any data D) Learning using images only **Answer:** B

10. What is the main role of an agent in an Artificial Intelligence system interacting with an environment?

A) Store data B) Make decisions and perform actions

C) Compile programs D) Display results **Answer:** B

11. Which library is widely used for Natural Language Processing tasks in Python programming?

A) NumPy B) NLTK

C) Pandas D) OpenCV **Answer:** B

12. What is the primary function of stopword removal during text preprocessing in NLP tasks?

A) Remove important keywords B) Remove common words like "is", "the"

C) Add extra words D) Encrypt text **Answer:** B

13. Which concept in game theory refers to a situation where no player benefits by changing their strategy alone?

A) Zero-sum game B) Nash Equilibrium

C) Heuristic D) Tokenization **Answer:** B

14. What is the function of the open list in the A search algorithm used in pathfinding problems?

A) Store explored nodes B) Store nodes to be explored

C) Store final path D) Store heuristic values **Answer:** B

15. What is the purpose of Optical Character Recognition (OCR) in AI-based systems and applications?

A) Audio processing B) Text extraction from images

C) Video editing D) Networking **Answer:** B

16. In reinforcement learning, what does a reward signal represent when an agent interacts with an environment and takes actions?

A) Punishment for incorrect actions B) Feedback indicating success or failure

C) Cost of computation D) Random output **Answer:** B

17. What is the key difference between supervised learning and reinforcement learning in terms of how feedback is provided to the model?

A) Supervised learning uses rewards

B) Reinforcement learning uses labeled data

C) Reinforcement learning uses interaction-based feedback

D) No difference

Answer: C

18. In Natural Language Processing, what is the role of tokenization when preparing raw text data for further processing?

A) Encrypt text B) Split text into smaller units like words

C) Store data D) Compress files **Answer:** B

19. What is the purpose of cosine similarity when comparing two documents in AI-based answer evaluation systems?

A) Measure distance B) Measure similarity between text vectors

C) Sort documents D) Encrypt data **Answer:** B

20. Which technique is used to extract important keywords from a dataset by considering both frequency and uniqueness of terms?

A) Sorting B) TF-IDF

C) Searching D) Hashing **Answer:** B

21. What is the role of the heuristic function in the A search algorithm when finding the shortest path in a graph?

A) Exact cost calculation B) Estimated cost to goal

C) Random value D) Final output **Answer:** B

22. In the A search algorithm, what is the significance of combining path cost and heuristic value for node evaluation?

A) Random search B) Optimal path finding

C) Data storage D) Encryption **Answer:** B

23. In game theory, what does a zero-sum game indicate about the relationship between players' gains and losses?

A) All players win B) One player's gain equals another's loss

C) No loss D) Random outcome **Answer:** B

24. What is Nash Equilibrium in a competitive game scenario where multiple players choose strategies independently?

A) Best mutual strategy B) Random decision

C) Worst case D) No solution **Answer:** A

25. In an AI-based answer evaluation system, how is similarity between student answer and model answer typically calculated?

A) Sorting B) Cosine similarity

C) Searching D) Encoding **Answer:** B

26. What is overfitting in machine learning models when trained on a limited dataset with high accuracy?

A) Poor performance B) Model fits training data too well

C) No learning D) Random output **Answer:** B

27. What is underfitting in machine learning when a model fails to capture patterns from the training data?

A) Perfect model B) Poor learning

C) High accuracy D) Complex model **Answer:** B

28. In NLP preprocessing, why is stopword removal important when analyzing large text datasets?

A) Increase noise B) Reduce unnecessary common words

C) Add complexity D) Encrypt data **Answer:** B